fer-cross-dataset

Implementation code for **"Beyond Single-Dataset Performance: A Controlled Benchmark and Diagnostic Study of Cross-Dataset Generalization in Facial Expression Recognition"** (SPIE Proceedings, 2026).

Overview

This repository provides a reproducible benchmark for cross-dataset generalization in Facial Expression Recognition (FER). Eight methods spanning four methodological categories are evaluated under a leave-one-dataset-out protocol across five FER datasets, with 10 independent training seeds per method for statistical reliability.

Environment

- Python 3.10+, PyTorch 2.x (CUDA recommended)
- Install: `pip install torch torchvision numpy scipy matplotlib pillow pywt`

Datasets

Five publicly available FER datasets. Download from official sources and place under `data/`:

Dataset	URL	Access
AffectNet	https://mohammadmahoor.com/pages/databases/affectnet/	Registration + license
RAF-DB	http://whdeng.cn/RAF/model1.html	Email application
FER2013	https://www.kaggle.com/datasets/msambare/fer2013	Kaggle account
CK+	https://zenodo.org/records/11221351	Open access
JAFFE	https://zenodo.org/records/14974867	Open access

Methods Evaluated

Category	Method	Backbone	Seeds
Baseline	ResNet-18	ResNet-18 (ImageNet)	10
FER-specific	SCN (Self-Cure Network)	ResNet-18	10
FER-specific	RUL (Relative Uncertainty Learning)	ResNet-18	10
FER-specific	MHAN (Multi-Head Hybrid Attention)	MixedFeatureNet (MS-Celeb-1M)	10
Data augmentation	RandAugment	ResNet-18	10

Category	Method	Backbone	Seeds
Data augmentation	MixUp	ResNet-18	10
Transformer	ViT-B/16	ViT-B/16 (ImageNet-21k)	10
Domain generalization	SWAD	ResNet-18	10

Random Seeds

All methods trained with 10 random seeds: **10, 42, 89, 123, 456, 781, 789, 999, 1337, 2026**.

Repository Structure

```
fer-cross-dataset/
├── src/                    # Core library
│   ├── dataset_registry.py # Unified dataset interface
│   ├── dataset_*.py        # Per-dataset loaders
│   ├── models.py           # Backbone builders
│   ├── iresnet.py          # ArcFace iresnet18
│   ├── scn_model.py        # SCN with rank regularization
│   ├── rul_model.py        # RUL with feature mixup
│   ├── wavelet.py          # Configurable DWT decomposition
│   ├── preprocess.py       # Face detection / preprocessing
│   ├── train.py            # Training loop
│   └── color_palette.py    # Unified color palette
├── scripts/                # Training and analysis scripts
│   ├── run_final_10seeds.py     # Main 10-seed training (all methods)
│   ├── run_all_seeds.py         # Batch training for core methods
│   ├── run_scn_10seeds.py       # SCN-specific training
│   ├── run_rul_10seeds.py       # RUL-specific training
│   ├── run_swad_10seeds.py      # SWAD training
│   ├── run_mhan_vit_fix.py      # MHAN + ViT-B training
│   ├── run_ablation.py          # Face pretraining ablation
│   ├── wavelet_sensitivity.py   # 7-wavelet sensitivity analysis
│   ├── multi_factor_analysis.py # 6-feature Spearman correlation
│   ├── scn_alpha_posthoc.py     # SCN alpha post-hoc analysis
│   ├── generate_figures.py      # Revision figures (5 new)
│   ├── redraw_fig1_fig2.py      # Updated Fig 1 + Fig 2
│   ├── frequency_signature_analysis.py # DWT frequency signature
│   ├── freq_generalization_cross.py    # Frequency-generalization correlation
│   ├── scn_alpha_analysis.py     # SCN alpha weight analysis
│   ├── bootstrap_ci.py           # Bootstrap confidence intervals
│   ├── statistical_tests.py      # Pairwise statistical comparisons
│   ├── analysis_figures.py       # Original figures (Fig 1-3)
│   └── tsne_analysis.py          # t-SNE feature visualization
├── runs/                   # Experiment outputs
│   ├── all_10seeds/       # Main 10-seed results
│   ├── wavelet_sensitivity/    # Wavelet analysis results
│   ├── multi_factor/           # Multi-factor Spearman results
```

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│   ├── scn_alpha_posthoc/      # SCN alpha post-hoc results
│   └── figures/          # Generated figures
└── pretrained/           # Pretrained weights
    └── ms1mv3_arcface_r18.pth  # MS1MV3 ArcFace
```

Quick Start

1. Main 10-seed training (all 8 methods):

```
python scripts/run_final_10seeds.py
```

2. Analysis scripts:

```
# Wavelet sensitivity (7 bases)
python scripts/wavelet_sensitivity.py

# Multi-factor dataset analysis
python scripts/multi_factor_analysis.py

# SCN alpha post-hoc
python scripts/scn_alpha_posthoc.py

# Frequency signature analysis
python scripts/frequency_signature_analysis.py
```

3. Generate revision figures:

```
# New figures (full ranking, wavelet, SCN alpha, multi-factor, MHAN vs ResNet)
python scripts/generate_figures.py

# Updated Fig 1 (heatmap) + Fig 2 (bar chart) with 8 methods
python scripts/redraw_fig1_fig2.py
```

Key Results (10-Seed, RAF-DB Source)

Method	Mean Macro-F1	Std	CV	p vs ResNet	Cohen's d
SWAD	0.289	0.009	3.2%	0.001	2.24
MHAN	0.283	0.024	8.6%	0.011	1.35
MixUp	0.272	0.020	7.5%	0.056	1.01
RUL	0.272	0.016	5.9%	0.040	1.10

Method	Mean Macro-F1	Std	CV	p vs ResNet	Cohen's d
SCN	0.270	0.010	3.6%	0.049	1.11
RandAug	0.264	0.022	8.5%	0.212	0.63
ViT-B/16	0.264	0.066	25.1%	0.594	0.25
ResNet-18	0.252	0.022	8.6%	—	—

Statistical significance assessed via Welch's t-test. Complete pairwise comparisons in the manuscript (Table 3).

Diagnostic Findings

- **Frequency-generalization correlation**: $r = 0.70$ ($p = 0.004$, $n = 15$), robust across 7 wavelet bases (all $p < 0.05$)
- **SCN alpha domain-shift detection**: 8.0% mean alpha reduction on cross-domain samples; above-median alpha samples achieve +0.080 accuracy delta
- **Training stability**: CNN methods CV 3.2-8.6%; ViT-B/16 CV 25.1%

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